

Kenny Mustapha

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EDUCATION

Simon Fraser University

Bachelor of Science in Computer Science

Burnaby, BC

Expected December 2027

EXPERIENCE

Student Software Engineer

May 2026 – Aug 2026

Develop for Good

Remote

- Developed a Next.js and PostgreSQL marketplace for Ikhuaya Lethu Care for the Elderly, giving community vendors an online storefront to support funding for elderly care programs.
- Translated client requirements into delivered marketplace features and implemented validation throughout vendor and storefront workflows to strengthen data integrity and reliability.

Software Engineering Intern

Jan 2026 – Apr 2026

Lillup

Remote

- Enabled audio generation from model outputs by integrating speech-generation components and implementing data conversion between model stages.
- Reduced experiment turnaround time by approximately 15% by parallelizing Python benchmarking pipelines and automating latency and audio-quality comparisons across multiple model configurations.
- Reduced manual validation effort by approximately 20% by building automated tests to compare model outputs across numerical precision settings.

Machine Learning Research Assistant

Jan 2026 – Apr 2026

SFU School of Computing Science

Burnaby, BC

- Built a Python pipeline combining YOLOv8, OpenCV camera calibration, and pinhole geometry to estimate object distance from monocular images and known object dimensions.
- Developed experimental workflows to benchmark depth estimates against physical measurements and analyze errors caused by camera parameters, object geometry, and detection bounding boxes.
- Served structured JSON experiment results through FastAPI to support reproducible experiments and downstream integration.

Software Engineering Intern

Sep 2025 – Dec 2025

Deepiri

Remote

- Reduced repeated database queries by approximately 30% across cacheable API routes by profiling PostgreSQL-backed endpoints under concurrent load and adding Redis caching and connection pooling.
- Created Python monitoring pipelines for request latency, cache hit rates, and connection pool utilization to help engineers diagnose backend bottlenecks during production incidents.
- Used profiling results to prioritize backend improvements with engineers, strengthening API reliability and performance during traffic spikes.

PROJECTS

LLM API Gateway | Go, PostgreSQL, Redis, Docker

- Developed a self-hosted API gateway in Go that unifies access to OpenAI, Anthropic, and Cohere, with model routing, rate limits, and live cost tracking.
- Enforced per-key quotas with a Redis-backed token bucket and recorded model usage, token counts, latency, and request costs in a PostgreSQL ledger.
- Implemented fallback routing to retry failed requests with secondary providers without changes to client integrations.

IntakeIQ | AWS, TypeScript, Python, Redis, Docker, Kubernetes

- Built a patient-intake and triage platform with TypeScript and Python on AWS, separating intake submission from asynchronous processing through backend microservices.
- Reduced processing latency by 40% through multithreaded execution and Redis caching, supporting real-time triage workflows while maintaining consistent patient data.
- Automated build and deployment workflows for containerized services using Docker and Kubernetes, adding AWS CloudWatch monitoring to diagnose failures and track service health.

TECHNICAL SKILLS

Languages: Python, TypeScript/JavaScript, Go, Java, SQL, C++

Frameworks & Libraries: FastAPI, Node.js, Express.js, Next.js, React, React Native

Databases & Developer Tools: PostgreSQL, Redis, MongoDB, AWS (EC2, S3), Prisma, Git, Docker, Codex, Claude Code

AI/ML: PyTorch, scikit-learn, OpenCV, YOLO, MediaPipe, NumPy, Pandas, pgvector